I did some notes for myself in the sheets then forgot to rewrite to a new sheets, sorry.

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TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
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STUDENT ID NUMBER: U2210251 | TION NUMBER IN THIS COURSE: 003

HW2 – SPRING 2024

COURSE NAME:	*Computer Architecture*
COURSE NUMBER:	SOC2060
EXAMINATION DATE:	08.05.2024 TIME: 22:54
EXAMINATION DURATION:	
ADDITIONAL MATERIALS ALLOWED TO USE:	None
SPECIAL INSTRUCTIONS:	*- Answers will only be evaluated if they are readable.* *- Show all your work. For some questions, you may get partial credit even if the end result is wrong due to a calculation mistake. If you make assumptions, state your assumptions clearly and precisely.*

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

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1.[10 points] ISA and ISA Trade-offs

(a)[10 points] True/False questions - Circle T if the statement is true, circle F otherwise:

In MIPS, condition codes are set automatically when a general purpose register is written: T / F

The performance of a Multi-cycle MIPS is always better than a Single-cycle MIPS: T / F

In Multi-cycle MIPS, intermediate states are recorded in PC and memory : T / F

The behavior of the entire processor is specified fully by a finite state machine: T / F

In Multi-cycle MIPS, control signals for the next state are determined in current state: T / F

In Single-cycle MIPS, the most complex operation slows down everything: T / F

By adopting pipelining, more concurrency can be achieved: T / F

In Multi-cycle MIPS, most of the datapath is idle when a memory access is happening: T / F

The resource contention can be avoided by adopting pipelining: T / F

In Single-cycle MIPS, the memory contains both instructions and data: T / F

2. [10 points] Single-Cycle Processor Datapath

Below is the picture of a single-cycle MIPS processor.

And the formats of instructions are as follows

31 26	25 21	20 16	15 11	10 6	5 0	
0	rs	rt	rd	shamt	funct	R-Type
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits	

31 26	25 21	20 16	15 0	
opcode	rs	rt	immediate	I-Type
6 bits	5 bits	5 bits	16 bits	

31 26	25 0	
opcode	immediate	J-Type
6 bits	26 bits	

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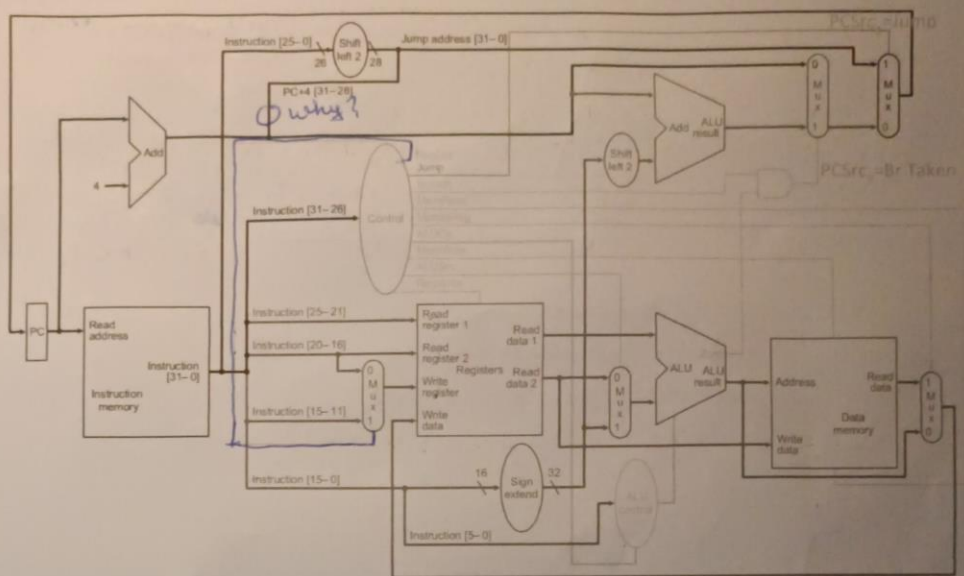
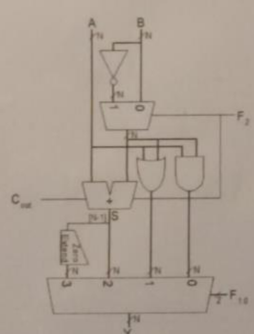


And ALUControl$_{2:0}$ is doing the same job as F$_{2:0}$ of following ALU Design.



~~add~~

000	A & B
001	A | B
010	A + B
011	not used
100	A & ~B
101	A | ~B
110	A − B
111	SLT

SLT - set if less than

(a)[10 points] A new R-type instruction "lw2" needs to be added to the instruction set. The semantics of "lw2" is as follows:

lw2: Rd <- Memory[Rs + Rt]

PC <- PC + 4

1 lw - load word

lw $1, 100($2)

$1 = M[$2 + 100]

~~$rt $rs imm~~

rt = M[rs + imm]

2 as the new lw2 doesn't use immidiate values anymory it's now R-type

rd = M[rs + rt]

3 ALL the control signals are utilized except Jump, Branch, MemW

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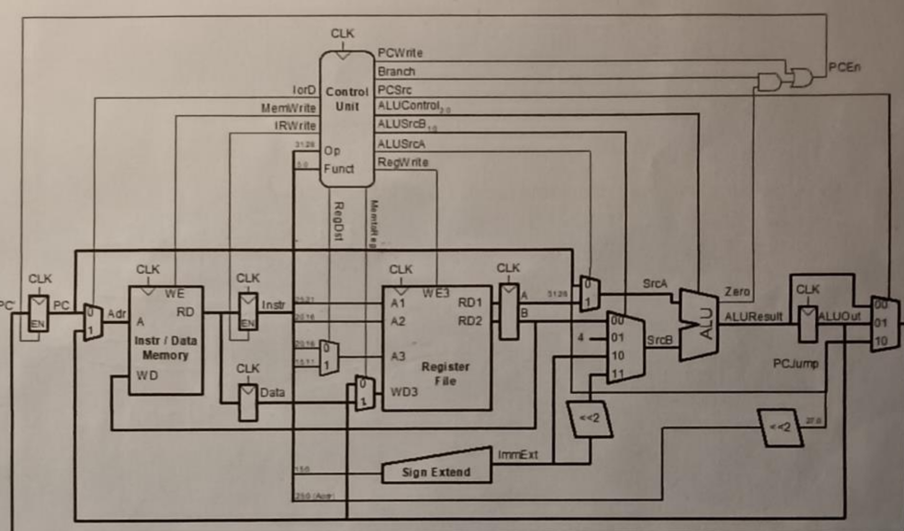
Control Signal	Value
RegDst	1
$PCSrc_1$	0
Branch	0
MemRead	1
MemtoReg	1
$ALUControl_{2:0}$	010
MemWrite	0
ALUSrc	0
RegWrite	1
$PCSrc_2$	0

it's 1 as we have rd
0 as we don't ~~branch~~ jumps
0 as we don't branch
1 as we read from mem
1 as we give the value from mem to reg
010 is func code for add
don't write to mem
0 as ~~rs~~ rt is used not imm
1 as data is to be written in a reg
0 as ~~jump~~ is not taken br.

3.[40 points] Multi-Cycle Processor Datapath

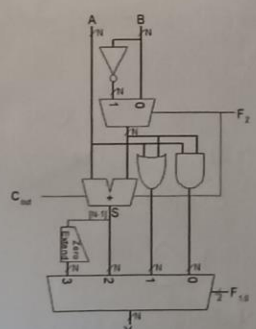
Below is the picture of a multi-cycle MIPS processor.

And instruction formats are the same as question 2.



And $ALUControl_{2:0}$ is doing the same job as $F_{2:0}$ in the following ALU Design.

Same as previous ALU



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(a)[10 points] List all the values of control signals at "write back" phase of LW (Load Word) instruction. If a control signal doesn't affect the "write back", write X meaning "don't care".

Control Signal	Value
IorD	X
RegWrite	1
IRWrite	0
RegDst	0
MemtoReg	1
PCWrite	0
$PCSrc_{1:0}$	XX
$ALUControl_{2:0}$	XXX
$ALUSrcB_{1:0}$	XX
ALUSrcA	X

rt = M[rs + imm]

(b)[10 points] List all the values of control signals at "memory access" phase of LW (Load Word) instruction. If a control signal doesn't affect the "memory access", write X meaning "don't care".

Control Signal	Value
IorD	1
RegWrite	0
IRWrite	0
RegDst	X
MemtoReg	X
PCWrite	0
$PCSrc_{1:0}$	XX
$ALUControl_{2:0}$	XXX
$ALUSrcB_{1:0}$	XX
ALUSrcA	X

(c)[10 points] List all the values of control signals at "decode / register operand fetch" phase of BRANCH instruction. If a control signal doesn't affect the "decode and register operand fetch", write X meaning "don't care".

Control Signal	Value
IorD	X
RegWrite	0
IRWrite	0
RegDst	X
MemtoReg	X
PCWrite	0
$PCSrc_{1:0}$	XX
$ALUControl_{2:0}$	010
$ALUSrcB_{1:0}$	11
ALUSrcA	0

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(d)[10 points] List all the values of control signals at "execute / address generation" phase of BRANCH instruction. If a control signal doesn't affect the "execute and address generation", write X meaning "don't care".

Control Signal	Value
IorD	X
RegWrite	0
IRWrite	0
RegDst	X
MemtoReg	X
PCWrite	0
$PCSrc_{1:0}$	01
$ALUControl_{2:0}$	110
$ALUSrcB_{1:0}$	00
ALUSrcA	1

4.[26 points] Caching

We have a byte addressable machine with L1, L2 cache and a memory. The size of a block is 16 bytes. The size of L1 cache is 512 bytes, the size of L2 cache is 2048 bytes, and the size of the memory is 2 giga bytes. The latency of L1 cache is 4 cycles, the latency of L2 cache is 8 cycles, and the latency of memory is 100 cycles. The L1 cache is a direct-mapped cache and the L2 cache is a fully associative cache. You are running a program which reads a byte from the memory 5 times. Assume both caches are empty and the memory has all the data which your instructions are accessing.

(a)[3 points] MM -> L2 -> L1 -> CPU

The address of the first byte to load is 0 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 112 cycles

Starting address(es) of block(s) in L1 cache: 0

Starting address(es) of block(s) in L2 cache: 0

MM -> L2 -> L1 -> CPU

(b)[5 points]

The address of the second byte to load is 16 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 112cycles

Starting address(es) of block(s) in L1 cache: 0,16

Starting address(es) of block(s) in L2 cache: 0,16

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(c)[6 points]

MM → L2 → L1 → CPU

The address of the second byte to load is 1025 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 112 cycles
Starting address(es) of block(s) in L1 cache: 16, 1024
Starting address(es) of block(s) in L2 cache: 0, 16, 1024

removed 0 from L1, as its DM cache

(d)[6 points]

L1 → CPU

The address of the second byte to load is 18 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 4 cycles
Starting address(es) of block(s) in L1 cache: 16, 1024
Starting address(es) of block(s) in L2 cache: 0, 16, 1024

nothing loaded found 16 in L1

(e)[6 points]

L2 → L1 → CPU

The address of the second byte to load is 15 and now the byte is loaded into a register. Specify how many cycles are needed for the load and specify the starting address(es) of block(s) of L1 and L2 caches.

The number of cycles needed: 12 Cycles
Starting address(es) of block(s) in L1 cache: 0, 16
Starting address(es) of block(s) in L2 cache: 0, 16, 1024

as 15 is inside 0 (0-15) loaded this instead of 1024

5.[14 points] Performance Evaluation

You need to choose a better processor to preform your experiment out of 2 processors A and B. The clock frequency of A is 600 MHz and the clock frequency of B is 700 MHz. Processor B has 3 more types of instructions (E,F,G), therefore the total number of executed instructions on B is 50% more than Processor A to perform your experiment.

Processor A

Instruction Class	CPI	Frequecy of Occurrece
A	2	40%
B	3	25%
C	3	25%
D	7	10%

Processor B

Instruction Class	CPI	Frequecy of Occurrece
A	2	15%
B	2	15%
C	4	10%
D	6	10%
E	1	10%
F	2	20%
G	2	20%

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(a)[8points]. What is the CPI of each processor A and B? Show your work.

CPIA

$0,4 \cdot 2 + 0,25 \cdot 3 + 0,25 \cdot 3 + 0,1 \cdot 7 = 3.0$

CPIB

$0,15 \cdot 2 + 0,15 \cdot 2 + 0,1 \cdot 4 + 0,1 \cdot 6 + 0,1 \cdot 1 + 0,2 \cdot 2 + 0,2 \cdot 2 = 2,5$

(b)[6points]. Which processor are you going to choose? Show your work in terms of total execution time. On Processor A, the total number of instructions to perform your experiment is 1,000,000.

Processor A or B?:

Total Time on Processor A: $1{,}000{,}000 \cdot 3 / 600 \cdot 10^6 = 0{,}005$

Total Time on Processor B: $1{,}500{,}000 \cdot 2{,}5 / 700 \cdot 10^6 \approx 0{,}0054$